Notes on Comodule Dualities

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This set of notes is largely based on [1].

1 Preparations

Let k be a field, and let C be a (coaugmented, conilpotent) coalgebra with a comultiplication map Δ and a counit map ε. Additionally, we denote by μ the multiplication on an algebra A and p the action map of an A-module.

Let $\mathcal{A}$ be a DG-category of A-modules (or C-comodules, C-contramodules, to be defined) with all cones and shifts. Let $\mathrm{Hot}(\mathcal{A})$ be the homotopy category of $\mathcal{A}$, with morphisms being chain maps modulo chain homotopy. Given a short exact sequence

$$0 \longrightarrow L \longrightarrow M \longrightarrow N \longrightarrow 0 \quad \text{in } \mathcal{A},$$

its **totalization** $\mathrm{Tot}(L \to M \to N)$ is the total complex of the associated short exact bicomplex, well-defined up to isomorphism in $\mathrm{Hot}(\mathcal{A})$.

Definition 1.1 (Comodule, Coderived Category)**.** A **(right)** C**-comodule** is a k-vector space M with a k-linear coaction

$$\rho : M \to M \otimes_k C$$

satisfying:

(1) Coassociativity:

$$(\rho \otimes \mathrm{id}_C) \circ \rho = (\mathrm{id}_M \otimes \Delta) \circ \rho$$

(2) Counitality:

$$(\mathrm{id}_M \otimes \varepsilon) \circ \rho = \mathrm{id}_M.$$

An object in $\mathrm{Hot}(\mathbf{Comod}\text{-}C)$ is **coacyclic** if it belongs to the smallest thick triangulated subcategory

$$\mathrm{Acycl}^{\mathrm{co}}(\mathbf{Comod}\text{-}C) \subset (\mathrm{Hot}\,\mathbf{Comod}\text{-}C)$$

closed under totalizations of short exact triples and infinite direct sums.

The **coderived category** is the Verdier quotient

$$D^{\mathrm{co}}(C) := \mathrm{Hot}(\mathbf{Comod}\text{-}C)\big/\mathrm{Acycl}^{\mathrm{co}}(\mathbf{Comod}\text{-}C).$$

Equivalently, $D^{\mathrm{co}}(C)$ is the homotopy category of $\mathbf{Comod}\text{-}C$ when we view $\mathbf{Comod}\text{-}C$ as a model category under Positselski's model structure, where the weak equivalences are morphisms with coacyclic cone, and the cofibrations are injections.

Definition 1.2 (Contramodule, Contraderived category)**.** A **(right)** C**-contramodule** is a k-vector space with a k-linear contraaction

$$\pi : \mathrm{Hom}_k(C, P) \to P$$

satisfying:

(1) Contraassociativity:

$$\pi \circ \mathrm{Hom}_k(\Delta, P) = \pi \circ \mathrm{Hom}_k(C, \pi) : \mathrm{Hom}_k(C \otimes C, P) \cong \mathrm{Hom}_k(C, \mathrm{Hom}_k(C, P)) \to P$$

(Intuitively, composing along comultiplication is composing along iterated contraaction)

(2) Counitality:

$$\pi \circ \mathrm{Hom}_k(\varepsilon, P) = \mathrm{id}_P.$$

Remark. As vector spaces, $\mathrm{Hom}_k(C,P) \cong \prod_C P$, and $M \otimes C \cong M \otimes (\bigoplus_C k) \cong \bigoplus_C M$, which is the reason why contramodules work well with products and limits, while comodules work well with coproducts and colimits.

A C-contramodule is **contraacyclic** if it belongs to the smallest thick triangulated subcategory

$$\mathrm{Acycl}^{\mathrm{ctr}}(\mathbf{Ctrmod}\text{-}C) \subset \mathrm{Hot}(\mathbf{Ctrmod}\text{-}C)$$

closed under totalizations of short exact triples and infinite products. The **contraderived category** is

$$D^{\mathrm{ctr}}(C) := \mathrm{Hot}(\mathbf{Ctrmod}\text{-}C)/\mathrm{Acycl}^{\mathrm{ctr}}(\mathbf{Ctrmod}\text{-}C).$$

Equivalently, $D^{\mathrm{ctr}}(C)$ is the homotopy category of $\mathbf{Ctrmod}\text{-}C$ when we view $\mathbf{Ctrmod}\text{-}C$ as a model category where the weak equivalences are morphisms with contraacyclic cone, and the cofibrations are surjections.

Let $\mathbf{fd}(C)$ denote the full subcategory on the compact objects of $D^{\mathrm{co}}(C)$, and in $D^{\mathrm{co}}(C)$ an object is compact if and only if it is weakly equivalent to a finite dimensional C-comodule. Let $\mathbf{thick}_{\mathcal{C}}(S)$ denote the smallest full triangulated subcategory of $\mathcal{C}$ containing S and closed under direct summands, for $\mathcal{C}$ triangulated and S a set of objects.

Lemma 1.3. *Let C be conilpotent.* $\mathbf{fd}(C) \simeq \mathbf{thick}_{D^{\mathrm{co}}}(k)$.

Proof. $\mathbf{thick}_{D^{\mathrm{co}}}(k) \subset \mathbf{fd}(C)$ is clear, since cones, shifts, and direct summands of a finite dimensional C-comodules is still finite dimensional.

For $\mathbf{fd}(C) \subset \mathbf{thick}_{D^{\mathrm{co}}}(k)$, it suffices to show that every finite dimensional C-comodules can be generated by k. For a conilpotent C, the coradical filtration

$$k = C^0 \subset C^1 \subset \cdots \subset C_n \subset \ldots$$

exhausts C. Then we can define an induced filtration for M:

$$M^i = \{m \in M : \rho(m) \in M \otimes C_i\}$$

and for finite dimensional M it has finite length. Note that $\rho : M^i \to M \otimes C_i$ induces a map

$$\tilde{\rho} : M^i/M^{i-1} \to (M/M^{i-1}) \otimes (C^i/C^{i-1}).$$

By considering the coassociativity property of ρ on M^i and quotienting out proper subspaces, we obtain $\tilde{\rho}(M^i/M^{i-1}) \subset (M/M^{i-1}) \otimes (C^0)$ i.e. the coassociativity equation shows that the image of $\tilde{\rho}$ lies in a subspace isomorphic to a C^0 tensor. Viewing $\tilde{\rho}$ as the new coaction, we prove M^i/M^{i-1} to be a C^0-comodule i.e. a k-comodule, which guarantees to be semisimple. Hence $M^i/M^{i-1} \cong k^n$ for some n.

Now for each i,

$$0 \longrightarrow M^{i-1} \longrightarrow M^i \longrightarrow k^{n_i} \longrightarrow 0$$

is a short exact sequence, and thus by taking cones (hence direct sums) and shifts we can obtain M^i from M^{i-1} and k. Inductively, any finite dimensional M is generated by k.

□

Theorem 1.4 (Finite dimensionality)**.** *Every C-comodule M is the filtered colimit (directed union) of its finite dimensional submodules. We write $M = \varinjlim_i M_i$.*

Proof. For tensor products, there exists finite collections of m_i and c_i such that

$$\rho(m) = \sum_{i=1}^{r} m_i \otimes c_i$$

where c_i's are linearly independent.

Consider the linear map

$$T : C^* \to M, \quad f \mapsto (\mathrm{id} \otimes f) \circ \rho.$$

Then clearly

$$T(f) = \sum_{i=1}^{r} m_i \otimes f(c_i), \quad N := \mathrm{im}(T) \subset \mathrm{span}\{m_1, \ldots, m_r\}.$$

Therefore N is finite dimensional, and $(\mathrm{id} \otimes \varepsilon)\rho(m) = m$ gives $m \in N$. By computing $\rho(n)$ any $n = (\mathrm{id} \otimes f)\rho(m)$ using map exchange and coassociativity, we have $n \in N \otimes C$. Therefore any $m \in M$ lies in such a finite dimensional subcomodule. Taking the union of M_i containing m over all $m \in M$, we obtain $M = \varinjlim_i M_i$.

2 Mod-Comod-Ctrmod Duality

Definition 2.1 (Injectives, Projectives, important bifunctors)**.** Let C-**Comod**-D be the category of C-D-bicomodules, and let C-**Ctrmod**-D be the category of C-D-bicontramodules. C° denotes the opposite coalgebra (a C°-comodule is a left C-comodule) and similarly for categories. Then there exists functors:

$$- \odot - : \textbf{Ctrmod}\text{-}C \times C\text{-}\textbf{Comod}\text{-}D \to \textbf{Comod}\text{-}D$$

$$\mathrm{Hom}_C(-,-) : (D\text{-}\textbf{Ctrmod}\text{-}C)^\circ \times \textbf{Comod}\text{-}C \to \textbf{Ctrmod}\text{-}D$$

$$\mathrm{Hom}^C(-,-) : (\textbf{Ctrmod}\text{-}C)^\circ \times D\text{-}\textbf{Ctrmod}\text{-}C \to D\text{-}\textbf{Ctrmod}$$

called **contratensor**, **comodule Hom**, and **contrahom** respectively, satisfying some associativity isomorphisms along with cotensor and Cohom.

Remark. (by Matt) "Hom" functors always land in the category of contramodules, aligned with their good compatibility with products and limits.

A C-comodule is **injective** if it is a direct summand of a **cofree** C-comodule $V \otimes C$ for some V, and A C-contramodule is **projective** if it is a direct summand of a **free** C-contramodule $\mathrm{Hom}(C, V)$ for some V. We denote the full subcategory of injective comodules and projective contramodules by $\mathbf{Inj}$-C and $\mathbf{Proj}$-C respectively.

Remark. $D^{\mathrm{co}}(C)$ is equivalent as a triangulated category to $\mathrm{Hot}(\mathbf{Inj}\text{-}C)$, the full triangulated subcategory of $\mathrm{Hot}(\mathbf{Comod}\text{-}C)$ consisting of injective C-comodules; $D^{\mathrm{ctr}}(C)$ is equivalent as a triangulated category to $\mathrm{Hot}(\mathbf{Proj}\text{-}C)$, the full triangulated subcategory of $\mathrm{Hot}(\mathbf{Ctrmod}\text{-}C)$ consisting of projective C-contramodules.

Proposition 2.2 (Homotopically well-behaved)**.** *For the functors above:*

$$- \odot - : \mathbf{Proj}\text{-}C \times C\text{-}\mathbf{Inj}\text{-}D \to \mathbf{Inj}\text{-}D$$

$$\mathrm{Hom}_C(-,-) : (D\text{-}\mathbf{Proj}\text{-}C)^\circ \times \mathbf{Inj}\text{-}C \to \mathbf{Proj}\text{-}D$$

$$\mathrm{Hom}^C(-,-) : (\mathbf{Proj}\text{-}C)^\circ \times D\text{-}\mathbf{Proj}\text{-}C \to D\text{-}\mathbf{Proj}$$

Proof. First, notice that all bifunctors are compatible with direct summands.

For contratensor products, if X is a projective C-contramodule and M is an injective C-D-comodule, $X \odot M$ is then a summand of $\mathrm{Hom}(C, V) \odot M \cong V \otimes M$. This isomorphism follows from the definition of contratensor products, and $X \odot M$ is a direct summand of a cofree C-comodule and hence injective.

For comodule Hom, if M is an injective D-C-comodule, N an injective C-comodule, then $\mathrm{Hom}_C(M, N)$ is a summand of a free D-contramodule:

$$\mathrm{Hom}_C(D \otimes W \otimes C, V \otimes C) \cong \mathrm{Hom}(D \otimes W \otimes C, V) \cong \mathrm{Hom}(D, \mathrm{Hom}(W \otimes C, V))$$

where the first isomorphism follows from definition. Hence $\mathrm{Hom}_C(M, N)$ is projective. For contrahom, we can also obtain a similar isomorphism from definition.

Note that Cohom and cotensor are also homotopically well-behaved.

□

Theorem 2.3 (Positselski)**.** *All the functors in 2.1 admit derived versions, and the derived versions of associativity isomorphisms hold.*

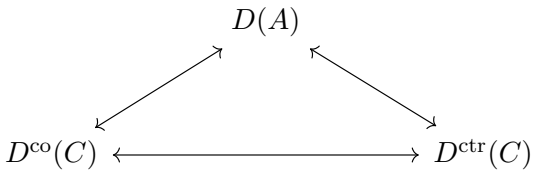
Intuitively, 2.2 ensures that derived versions of all the functors land in good subcategories and thus are well-defined. It also makes sure that the induced functors in 2.3 go from $D^{\mathrm{co}}(C)$ to $D^{\mathrm{ctr}}(C)$.

Theorem 2.4 (Co-contra correspondence)**.** *The functors $-\odot C$ and $Hom_C(C,-)$ form a Quillen equivalence between $\mathbf{Comod}$-C and $\mathbf{Ctrmod}$-C. The same pair of functors give a triangle equivalence $D^{\mathrm{co}}(C) \simeq D^{\mathrm{ctr}}(C)$.*

Remark. The correspondence holds since the given functors are their own derived functors, which results from the fact that $D^{\mathrm{co}}(C)$ is equivalent to the homotopy category of complexes of injective C-comodules, and $D^{\mathrm{ctr}}(C)$ is equivalent to the homotopy category of complexes of projective C-contramodule.

Now comes the main theorem. We say (A, C) is a **Koszul duality pair** if there is a fixed quasi-isomorphism $C \simeq BA$, which naturally defines a twisting cochain $\tau : C \to A$ (by bar-cobar conjunction from Loday-Vallette).

Theorem 2.5 (Mod-Comod-Ctrmod duality)**.** *Let (A, C) be a Koszul duality pair. There is a commutative diagram of quasi-equivalences of pretriangulated dg categories:*



where, given A-module M, C-comodule N, and C-contramodule X, the functors are:

- *a Quillen equivalence between $\mathbf{Mod}$-A and $\mathbf{Comod}$-C, given (induced) by the twisted tensor products $-\tilde{\otimes}C : \mathbf{Mod}\text{-}A \to \mathbf{Comod}\text{-}C$ and $-\tilde{\otimes}A : \mathbf{Comod}\text{-}C \to \mathbf{Mod}\text{-}A$ for coalgebra C and algebra A. The differentials are twisted by $(p \otimes id) \circ (id \otimes \tau \otimes id) \circ (id \otimes \Delta)$ and $(id \otimes \mu) \circ (id \otimes \tau \otimes id) \circ (\rho \otimes id)$ as usual,*
- *a Quillen equivalence between $\mathbf{Mod}$-A and $\mathbf{Ctrmod}$-C, given by the twisted Hom functors $\underset{\sim}{\mathrm{Hom}}(C,-) : \mathbf{Mod}\text{-}A \to \mathbf{Ctrmod}\text{-}C$ and $\underset{\sim}{\mathrm{Hom}}(A,-) : \mathbf{Comod}\text{-}C \to \mathbf{Mod}\text{-}A$ for coalgebra C and algebra A. The differential of $\underset{\sim}{\mathrm{Hom}}(C,M)$ is twisted by $p \circ (- \otimes \tau) \circ \Delta$ (i.e. the **convolution** with the twisting cochain $-*\tau$). The differential of $\underset{\sim}{\mathrm{Hom}}(A,X) : \mathbf{Comod}\text{-}C \to \mathbf{Mod}\text{-}A$ is twisted by $\pi \circ - \circ (\mu \circ (\tau \otimes id))$,*
- *the co-contra correspondence in 2.4.*

Moreover, there are the following correspondence of objects:

- *$A \in D(A)$ corresponds to $k \in D^{\mathrm{co}}(C)$,*
- *$k \in D(A)$ corresponds to $C \in D^{\mathrm{co}}(C)$ and $C^* \in D^{\mathrm{ctr}}(C)$,*
- *$A^* \in D(A)$ corresponds to $k \in D^{\mathrm{ctr}}(C)$.*

Remark. Notice that for any A-module M, $M\widetilde{\otimes}C$ is a cofree comodule with coaction map $\mathrm{id}\otimes\Delta$, and $\underset{\sim}{\mathrm{Hom}}(C,M)$ is a free contramodule with contraaction map Δ^*. Their injectivity/projectivity ensures the two equivalences are well-defined.

Also note that the twisted functors are similar to the twisted tensor products and and twisted homs for non-graded cases, as Loday-Vallette shows. Even $\underset{\sim}{\mathrm{Hom}}(A, M)$ is twisted by something like convolutions.

Remark. A dg functor $F : \mathcal{C} \to \mathcal{D}$ is a **quasi-equivalence** iff:

(1) **(Quasi-fully faithful)** For all objects X, Y the map of complexes

$$\mathrm{Hom}_{\mathcal{C}}(X, Y) \xrightarrow{F} \mathrm{Hom}_{\mathcal{D}}(FX, FY)$$

is a quasi-isomorphism.

(2) **(Homotopy-essentially surjective)** The induced functor on homotopy categories

$$H^0(F) : H^0(\mathcal{C}) \longrightarrow H^0(\mathcal{D})$$

is essentially surjective (every object of $H^0(\mathcal{D})$ is isomorphic to some $H^0(F)(X)$).

If $\mathcal{C}, \mathcal{D}$ are pretriangulated (so their homotopy categories are triangulated), any quasi-equivalence automatically induces a triangulated equivalence

$$H^0(\mathcal{C}) \simeq H^0(\mathcal{D}).$$

Moreover, a Quillen equivalence always restricts to a quasi-equivalence, as we have seen in 2.5.

Proof. We will only prove how Koszul duality is necessary for the functors to be equivalences, and in addition prove the correspondence of objects.

Let $F = -\tilde{\otimes}C$ and $G = -\tilde{\otimes}A$ in 2.5, Then:

$$F(G(N)) \cong N\tilde{\otimes}(A\tilde{\otimes}C), \quad G(F(M)) \cong M\tilde{\otimes}(C\tilde{\otimes}A)$$

By the twisting morphism fundamental theorem in Loday-Vallette, since (A, C) is a Koszul duality pair, $A\tilde{\otimes}C$ and $C\tilde{\otimes}A$ are both acyclic; indeed $A\tilde{\otimes}C \simeq k$ and $C\tilde{\otimes}A \simeq k$. Thus:

$$N\tilde{\otimes}(A\tilde{\otimes}C) \simeq N\tilde{\otimes}k \simeq N, \quad M\tilde{\otimes}(C\tilde{\otimes}A) \simeq M\tilde{\otimes}k \simeq M$$

and F, G are equivalences.

Similarly, for $F_{\mathrm{ctr}} = \underset{\sim}{\mathrm{Hom}}(C, -)$ and $G_{\mathrm{ctr}} = \underset{\sim}{\mathrm{Hom}}(A, -)$, we have:

$$F_{\mathrm{ctr}}G_{\mathrm{ctr}}(X) \simeq \underset{\sim}{\mathrm{Hom}}(C, \underset{\sim}{\mathrm{Hom}}(A, X)) \simeq \underset{\sim}{\mathrm{Hom}}(C\tilde{\otimes}A, X) \simeq \underset{\sim}{\mathrm{Hom}}(k, X) \simeq X, \quad G_{\mathrm{ctr}}F_{\mathrm{ctr}}(M) \simeq M$$

since the functors preserve quasi-isomorphisms by projectivity.

Under this correspondence, $A \mapsto A\tilde{\otimes}C \simeq k$ as above; $k \mapsto k\tilde{\otimes}C \simeq C \in D^{\mathrm{co}}(C)$, and $k \mapsto \underset{\sim}{\mathrm{Hom}}(C, k) = C^* \in D^{\mathrm{ctr}}(C)$; $A^* \mapsto \underset{\sim}{\mathrm{Hom}}(C, \mathrm{Hom}(A, k)) \simeq \underset{\sim}{\mathrm{Hom}}(C \otimes A, k) \simeq \underset{\sim}{\mathrm{Hom}}(k, k) \simeq k$.

□

Corollary 2.6. *For Koszul duality pair (A, C), there are algebra quasi-isomorphisms:*

$$A \simeq \mathbb{R}\mathrm{End}_{D^{\mathrm{co}}(C)}(k)$$
$$\mathbb{R}\mathrm{End}_A(A^*) \simeq \mathbb{R}\mathrm{End}_{D^{\mathrm{ctr}}(C)}(k)$$
$$C^* \simeq \mathbb{R}\mathrm{End}_A(k)$$

Proof. Since the functors in 2.5 are quasi-equivalences (which induce quasi-isomorphisms on derived morphisms), we have

$$A \simeq \mathrm{Hom}_A(A,A) \simeq \mathbb{R}\mathrm{End}_A(A) \simeq \mathbb{R}\mathrm{End}_{D^{\mathrm{co}}(C)}(k)$$

$$\mathbb{R}\mathrm{End}_A(A) \simeq \mathbb{R}\mathrm{End}_{D^{\mathrm{ctr}}(C)}(k)$$

$$C^* \simeq \mathrm{Hom}_C(C,C) \simeq \mathbb{R}\mathrm{End}_{D^{\mathrm{co}}(C)}(C) \simeq \mathbb{R}\mathrm{End}_A(k)$$

where $C^* \simeq \mathrm{Hom}_C(C,C)$ since every $f \in \mathrm{Hom}_C(C,C)$ is determined by $\varepsilon \circ f$, while the inverse is given by something similar to T in the proof of 1.4. □

3 Linear duals for comodules

Lemma 3.1 (Keller)**.** *For a pretriangulated category $\mathcal{T}$ with a compact generator G and a pretriangulated category $\mathcal{U}$, any exact dg functor $F : \mathcal{T} \to \mathcal{U}$ is determined on* $\mathbf{thick}(k)$ *by:*

- *the image $F(G)$*
- *the induced morphism of dg algebras $\mathbb{R}\mathrm{End}_{\mathcal{T}}(G) \to \mathbb{R}\mathrm{End}_{\mathcal{U}}(H(G))$.*

Intuitively, these two determine the functor's behavior on all the objects and morphisms between them.

Proposition 3.2. *For Koszul duality pair (A, C), there is a commutative diagram of functors*

$$\begin{array}{ccc} D(A)^\circ & \xrightarrow{\mathbb{R}\mathrm{Hom}_A(-,A)} & D(A^\circ) \\ \simeq \downarrow & & \downarrow \simeq \\ D^{\mathrm{co}}(C)^\circ & \xrightarrow{(-)^*} & D^{\mathrm{co}}(C^\circ) \end{array}$$

Moreover, the functor $(-)^*$ can be defined as follows. For any C-comodule M, write $M = \varinjlim_i M_i$ as in 1.4, and $M^* := \varprojlim_i M_i^*$.

Proof. Consider the composition:

$$F : D^{\mathrm{co}}(C)^\circ \xrightarrow{\simeq} D(A)^\circ \xrightarrow{\mathbb{R}\mathrm{Hom}_A(-,A)} D(A^\circ) \xrightarrow{\simeq} D^{\mathrm{co}}(C^\circ).$$

We first show $F = (-)^*$ at k. Since $\mathbb{R}\mathrm{Hom}_A(A, A) = \mathrm{Hom}(A, A) \cong A$,

$$F : k \mapsto A \mapsto A \mapsto k$$

by 2.5. Thus $F(k) \cong k \cong k^*$ for one-dimensional k.

Then by 2.6, $\mathbb{R}\mathrm{End}_{D^{\mathrm{co}}(C)}(k) \simeq A$ and $\mathbb{R}\mathrm{End}_{D^{\mathrm{co}}(C^\circ)}(k) \simeq A^\circ$. Across the Koszul duality, it can be checked that the map induced by F, $F_{k,k} : \mathbb{R}\mathrm{End}_{D^{\mathrm{co}}(C)}(k) \to \mathbb{R}\mathrm{End}_{D^{\mathrm{co}}(C^\circ)}(k)$ is identified with id_A, which is $(-)^*_{k,k}$. Hence $F_{k,k} \simeq (-)^*_{k,k}$ on $\mathbb{R}\mathrm{End}_{D^{\mathrm{co}}(C)}(k)$, and applying 3.1 with $G = k$, we obtain $F = (-)^*$ on $\mathbf{thick}_{D^{\mathrm{co}}(C)}(k)$. Thus by 1.3, $F = (-)^*$ on the subcategory of finite dimensional objects $\mathbf{fd}(C) \simeq \mathbf{thick}_{D^{\mathrm{co}}(C)}(k)$.

It remains to extend the argument for arbitrary M. By 1.4, $M = \varinjlim_i M_i$ for any M. F and $\mathbb{R}\mathrm{Hom}_A(-, A)$ are right derived and thus preserve homotopy limits, turning colimits into homotopy limits when passing to the opposite category. This means that

$$F(M) = F(\varinjlim_i M_i) \simeq \varprojlim_i F(M_i) \simeq \varprojlim_i M_i^* = M^*,$$

which is how we define M^* for any M.

□

The basic idea is to prove the statement for k (and derived endomorphisms), then any finite dimensional C-comodule, and finally any C-comodule.

4 One-sided dual for contramodules

Theorem 4.1. *For (A, C) a Koszul duality pair, there is a commutative diagram:*

$$\begin{array}{ccc} D^{\mathrm{ctr}}(C)^\circ & \xrightarrow{\mathbb{R}\mathrm{Hom}^C(-,C^*)} & D^{\mathrm{ctr}}(C^\circ) \\ \simeq\downarrow & & \downarrow\simeq \\ D(A)^\circ & \xrightarrow{M\mapsto M^*} & D(A^\circ) \end{array}$$

Proof. The idea is to directly prove the weak equivalence:

$$\mathbb{R}\mathrm{Hom}^C(\underset{\sim}{\mathrm{Hom}}(C,M),C^*)\simeq \underset{\sim}{\mathrm{Hom}}(C,M^*)$$

starting from $D(A)^\circ$.

This comes in several steps:

- Since $\underset{\sim}{\mathrm{Hom}}(C,M)$ is projective (free),

$$\mathbb{R}\mathrm{Hom}^C(\underset{\sim}{\mathrm{Hom}}(C,M),C^*)\simeq \mathrm{Hom}^C(\underset{\sim}{\mathrm{Hom}}(C,M),C^*)$$

- There is a natural contramodule adjunction $\mathrm{Hom}^C(\mathrm{Hom}(C,V),X)\to\mathrm{Hom}(V,X)$ for any vector space V and contramodule X, sending F to the function $v\mapsto F(v\eta)$; this is basically restoring how F acts on V. Then with Hom-tensor adjunction we have:

$$\mathrm{Hom}^C(\underset{\sim}{\mathrm{Hom}}(C,M),C^*)\cong\mathrm{Hom}(M,C^*)\cong(M\otimes C)^*\cong(C\otimes M)^*\cong\mathrm{Hom}(C,M^*).$$

- Now there is an isomorphism on the underlying vector spaces

$$\Xi:\mathbb{R}\mathrm{Hom}^C(\underset{\sim}{\mathrm{Hom}}(C,M),C^*)^\#\to\underset{\sim}{\mathrm{Hom}}(C,M^*)^\#$$

  and it remains to check that Ξ is compatible with differentials. Denote $F\phi$ and ψ the differentials on the source and target (on hom complex the differential is indeed $F\phi$, where ϕ is the differential on the free contramodule). This follows from direct calculations.

□

References

[1] M. Booth, J. Chuang, and A. Lazarev. Nonsmooth Calabi–Yau structures for algebras and coalgebras, 2025. URL https://arxiv.org/abs/2502.12162.